

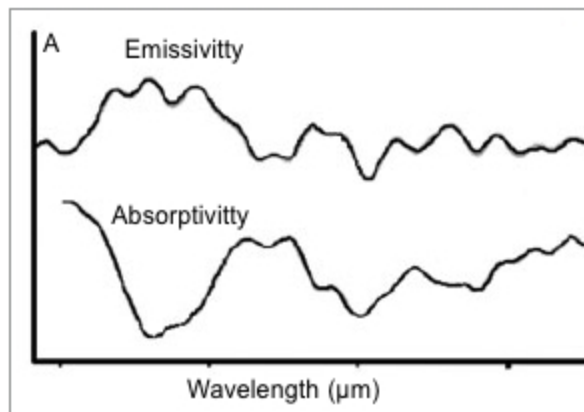


Fig. 7: Absorptivity and emissivity (arbitrary units/au) of 220mg/dl glucose in KBr sample at 41°C

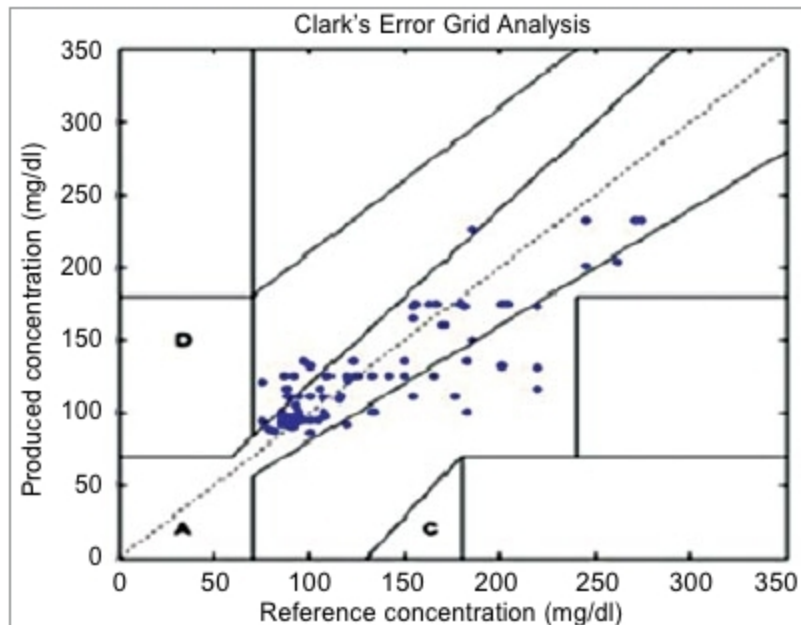


Fig. 8: Clarke's plot of EGA - NIR spectroscopy

ments gives a cumulative correlation between invasive (laboratory) and non-invasive glucose concentrations, assuming error-less invasive glucose measurements. In Clarke EGA, clinically-accurate zone A contains 81 per cent of non-invasive data, and clinically-acceptable zone B contains 19 per cent data. One data point was close to zone D, as a clinically-significant error. This experiment presents non-invasive concept validity by showing good correlation between NIR transmittance and BG.

3. IR glucose monitoring is based on the spectral characteristics of various constituents of blood, which are separated using analytical chemistry spectroscopy methods. The instrument relies on the use of IR filters and IR detector windows. One filter passes radiation through thermal emission bands with glucose signatures and is placed in one of the IR detector windows. Comparison of radiation intensity between the two detector windows provides measurement proportional to the analyte concentration that can be correlated with the concentration of BG.

4. Clinical studies were performed using a new and advanced handheld prototype of a non-invasive BG monitor based on thermal emission spectroscopy (TES) IR spectro-radiometer. Patients with type 1 and type 2 diabetes were tested using this device. The subject's sample size was determined by power analysis and evaluated for validation in a complete predictive mode for patients and operators. Measurements demonstrated the reproducibility of the methodology of the non-invasive glucose monitor.

Statistical analysis of non-invasive data was performed using error-in-variables methods, namely, Deming Passing, Bablok and orthogonal regression with parameter estimation. Results from Passing and Bablok methods of comparison for slope and intercept of linear regression are comparable with orthogonal regression with other statistical parameter estimations. Clarke EGA showed that 100 per cent of the measurements fell within zones A and B. In clinically-accurate zone A, there was 90 per cent non-invasive data. In clinically-acceptable zone B, there was 10 per cent data.

TES is a truly non-invasive prototype based on thermal emission in the mid-IR spectral region. It has shown glucose measurements with clinically-acceptable accuracies without individual daily calibrations.

Thermal IR-emission spectra from plasma at temperatures with different glucose concentrations can be plotted to see the peak intensities of deconvoluted spectral bands (arbitrary units/au). This is shown in Fig. 7.

Errors in laboratory measurements as an additive quantity are increasing the real errors of non-invasive results. For comparison with another non-invasive monitor, one can refer to data reported for US FDA-approved (March 2001) GlucoWatch system (Cygnus). This requires a three-hour warm-up time and individual calibration every

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